Studying toxic effects of Alzheimer's risk variant in *CASP8* using cellular models

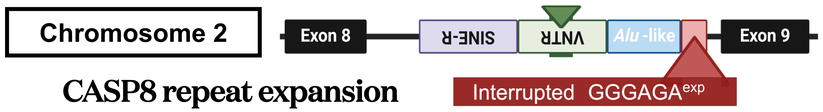
Ryan Xiao[12345], Denise Romero[1234], Rodrigo F. Tomas[1234], Lien Nguyen[1234]

University of Florida, Gainesville, FL, USA[1]; UF Center for NeuroGenetics, Gainesville, FL, USA[2]; UF Department of Molecular Genetics & Microbiology, Gainesville, FL, USA[3]; Genetics Institute, Gainesville, FL, USA[4]; UF Student Science Training Program[5]

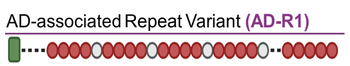
Background

- Alzheimer's Disease is a neurodegenerative disease impacting ~7 million Americans and is the most common form of dementia
- Characterized by accumulation of amyloid-β plaques (Aβ) and hyperphosphorylation of tau (pTau)
- **The genetics of Alzheimer's remains unknown**
- Our lab discovered a GGGAGA^{exp} repeat expansion in *CASP8* which encodes for poly-glycine-arginine-containing (polyGR) aggregates accumulating in AD brains
- Our lab has generated an adeno-associated virus (AAV) neuroblastoma (SH-SY5Y) cell model that expresses GGGAGA^{exp} to assess AD neuropathological changes *in vitro*

CASP8 repeat expansion linked to AD



CASP8 repeat expansion

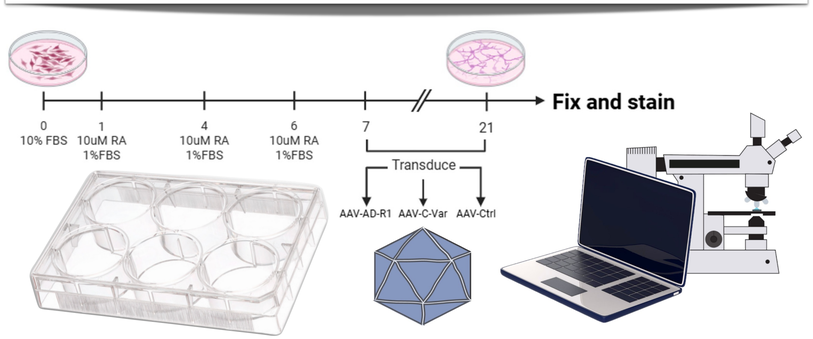


	CASP8 AD-R1	Other Patterns	Total Cases	Odds Ratio	95 % CI	Z Statistic	*p* Value
AD	88 (7.5%)	1086	1174	2.2	[1.5185, 3.31896]	4.166	3.1×10^{-5}
Control	44 (3.6%)	1195	1239				

PolyGR-encoding repeat expansion (AD-R1) was found to **significantly increase AD risk**

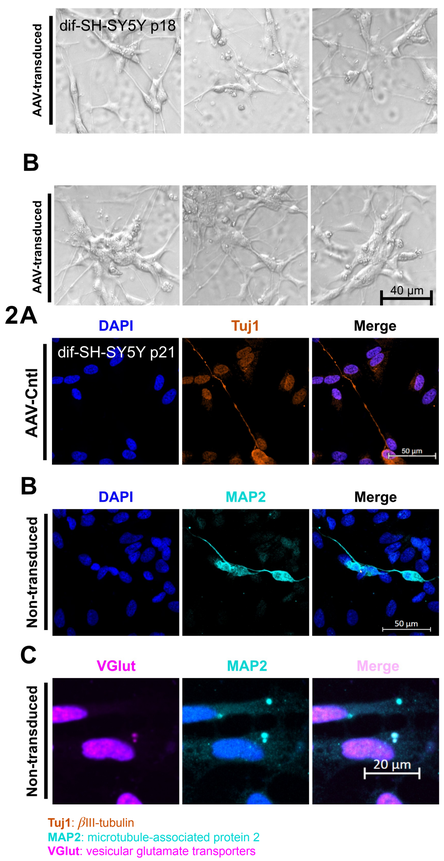
2.2 Odds Ratio

Experimental Design



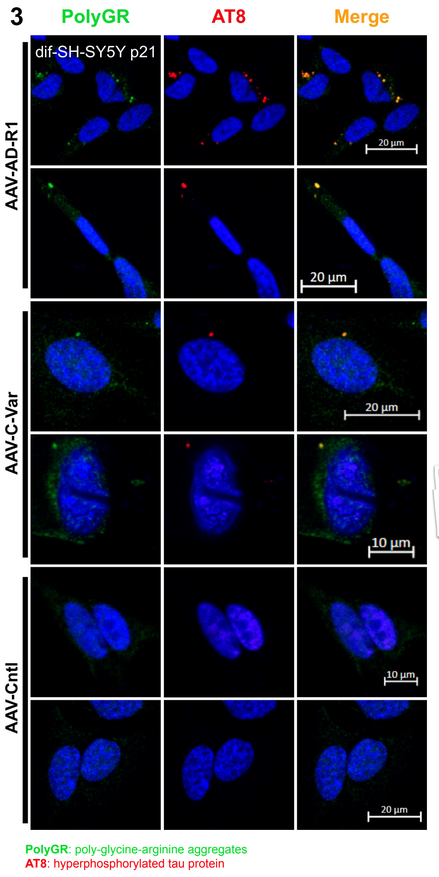
Results

Images confirm SH-SY5Y differentiation



Tuj1: βIII-tubulin
MAP2: microtubule-associated protein 2
VGlut: vesicular glutamate transporters

polyGR leads to increased pTau



PolyGR: poly-glycine-arginine aggregates
AT8: hyperphosphorylated tau protein

Conclusion

- SH-SY5Y cells transduced with C8-C-Var or C8-AD-R1 express polyGR
- PolyGR aggregates partially co-localize with pTau, suggesting polyGR accumulation causes increased pTau in differentiated SH-SY5Y cells
- **Our differentiated SH-SY5Y transduced with AAV expressing the *CASP8*-GGGAGA^{exp} can be used to model pathological changes related to AD**
- Future directions include investigation on Aβ expression, staining for different cell markers (astrocytes, oligodendrocytes, microglia, etc.) or neuron types (GABAergic, dopaminergic, cholinergic, etc.) and shapes (pyramidal, bipolar, unipolar, etc.), and different staining techniques (FISH, IHC, etc.) to study repeat RNA and other protein pathologies

Acknowledgements

Nguyen Lab

Dr. Lien Nguyen, Alissa Greene, Allison Lewis, Andrea Suarez, Arshia Sethi, Corben Dean, Denise Romero, Elizabeth Michels, Ella Klyce, Hua-Hsin Hsiao, Dr. Huong Phuong, Isabella Gerstin, Jansen Reece, Johnathan Poisson, Kar Men Lee, Logan Bell, Michael J. Barbarito, Rodrigo Tomas, Ross Ellwood, Stefani Ng